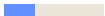


Environment-Specific Failure Modes in Default Merge

Environment	State dim	\# Actions	Dominant failure stage	Failed action mismatch	Mergeable crossing	Low-density signal	Interpretation
MountainCar2	2	3	Aggregation (mild) smaller drop than CartPole/LunarLander	0.62 	28% 	-	Same interval-collapse story, but milder than LunarLander.
CartPole	4	2	Support pruning hard tau prunes 21/25 groups	0.44 	30% 	73% midpoint 	Cleaner 2-action structure; many matched groups remain usable, but hard retention removes them early.
LunarLander8	8	4	Aggregation dominant loss appears before support	0.69 	65% 	0.48 bridge 	8D / 4 actions with minority-action difficulty; merged intervals compress multi-modal, action-inconsistent regions.

Transfer contrast: MiniGrid (14D, 3 actions, PPO) remains more stable despite higher dimension; the story is geometry + action semantics + merge behavior, not dimension alone.